

IK 556: Sūrya Siddhānta

Course Code	IK 556
Course Name	Sūrya Siddhānta
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	BTech/MTech/MS/MSc/MA/Ph.D
Mutual Exclusion	None
Approval	42nd Senate (Jan 2024)

Course Contents

- **Introduction to Sūrya Siddhānta** (12 Hours)
 - Historical Context and Origin. Overview of the key concepts.
 - Significance in Indian Astronomy.
 - Introduction to Indian calendrical computations.
- **Time Measurement and Planetary Models** (10 Hours)
 - Concepts of time: Yugas, Kalpas, Manvantaras.
 - Day and Night calculation, Solar and Lunar days.
 - Sidereal, Tropical, and Anomalistic months.
 - Planetary models: Epicycles and Eccentricities.
- **Mathematics in Sūrya Siddhānta** (10 Hours)
 - Trigonometric concepts: Sine, Cosine and R sine tables.
 - Mathematical techniques for astronomical computations.
 - Determining the positions of planets.
 - Eclipses: Calculation and Prediction.
- **Applications and Impacts** (10 Hours)
 - Influence on later astronomical texts and practices.
 - Comparison with other Siddhantas.
 - Contemporary relevance and applications.
 - Cross-cultural influences and exchanges with other astronomical traditions.

Text Books

1. Chakravarty, A. K., **The Suryasiddhanta: the astronomical principles of the text**, Asiatic Society, 2001.
2. Burgess, E., **Translation of the Surya Siddhanta: A Text-Book of Hindu Astronomy**, Yale University Press, 1860.

References

1. Gangooly, P. (Ed.), **The Sūrya Siddhānta: A textbook of Hindu astronomy**, Motilal Banarsidass Publishers, 1997.
2. **Digital Resource on Surya Siddhanta**: <https://www.wilbourhall.org/pdfs/suryaEnglish.pdf>